

SDG-BASED QUESTION – CO2

MACHINE LEARNING

**Compare Hypothesis Space Search in Genetic Algorithms versus
Gradient-Based Search in Backpropagation, Using Flood Prediction
Modeling as a Case Study**

SDG 13 – Climate Action

Course Code: ITA0609

Course Faculty : DR. Shri Vindhya A

Name : K.Vishwanathan

Register No.: 192312063

Department: Electronics and Communication Engineering

Academic Year: 2026

1. Introduction

Machine Learning provides several methods for searching for an effective model from a large set of possible solutions. Two important approaches are hypothesis space search using Genetic Algorithms and gradient-based search using Backpropagation. Although both methods aim to improve a model's performance, they explore the solution space in fundamentally different ways.

A Genetic Algorithm searches for promising solutions using evolutionary operations such as selection, crossover, and mutation. It treats possible models, parameters, rules, or feature combinations as candidate solutions and searches for a high-quality solution based on a fitness function.

Backpropagation, on the other hand, is a gradient-based optimization method used to train neural networks. It calculates the error between the predicted output and the actual output, determines how each weight contributed to the error, and updates the weights in the direction that reduces the loss.

Flood prediction is an appropriate case study because climate change has increased the importance of understanding extreme rainfall, river-level changes, soil moisture, and other environmental factors. A reliable flood prediction model can support early warning, disaster preparedness, and climate resilience.

Objectives

- To explain hypothesis space search using Genetic Algorithms.
- To explain gradient-based search using Backpropagation.
- To compare the search mechanisms, strengths, and limitations of both approaches.
- To apply both approaches to a flood prediction modeling scenario.
- To connect Machine Learning-based flood prediction with SDG 13 – Climate Action.

Case Study Problem

The objective of the flood prediction system is to estimate flood risk using environmental variables such as rainfall intensity, river water level, soil moisture, temperature, humidity, and historical flood information. The model may produce a binary output such as Flood/No Flood or a multi-class output such as Low, Medium, and High Flood Risk.

2. Flood Prediction Modeling as a Machine Learning Problem

A flood prediction model learns a relationship between environmental input variables and a target flood condition. Historical observations are used to train the model, while unseen observations are used to test its ability to generalize.

2.1 Input Features

Feature	Example	Role in Prediction
Rainfall Intensity	120 mm/hour	Indicates water accumulation
River Water Level	4.5 m	Shows river overflow risk
Soil Moisture	85%	Indicates reduced absorption capacity
Temperature	30°C	Provides environmental context

Humidity	90%	Supports weather condition analysis
Historical Flood Data	Previous flood events	Provides past pattern information

2.2 Target Output

The target output can be represented as:

- 0 = No Flood / Low Risk
- 1 = Flood / High Risk
- Alternatively: Low Risk, Medium Risk, and High Risk

2.3 General Modeling Workflow

The overall workflow is:

Historical Environmental Data → Data Preparation → Search/Optimization → Model Training →
Flood Risk Prediction

The main difference between the two approaches is the way they search for a suitable model. Genetic Algorithms search through a population of candidate hypotheses, whereas Backpropagation continuously adjusts neural-network weights using gradient information.

3. Hypothesis Space Search Using Genetic Algorithms

A hypothesis space is the collection of all possible hypotheses or candidate solutions that a learning system can consider. In a flood prediction problem, a hypothesis may represent a set of feature weights, decision rules, thresholds, model parameters, or even a complete model structure.

3.1 Representation of a Candidate Solution

For example, a candidate flood-risk model can be represented by parameters associated with rainfall, river level, and soil moisture:

$$\mathbf{H} = [\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \text{threshold}]$$

Each candidate solution represents one possible hypothesis about how environmental variables influence flood risk.

3.2 Genetic Algorithm Search Process

1. Initialization: Generate a population of random candidate solutions.
2. Fitness Evaluation: Test each candidate on historical flood data and calculate its prediction performance.
3. Selection: Choose candidates with better fitness values.
4. Crossover: Combine parts of selected candidates to create new solutions.
5. Mutation: Randomly alter parts of a candidate to introduce diversity.
6. Replacement: Form a new population using the better solutions.
7. Termination: Stop when a satisfactory fitness value or maximum number of generations is reached.

3.3 Fitness Function

The fitness function measures how well a candidate solution predicts flood events. A possible fitness function is:

$$\text{Fitness} = \text{Prediction Accuracy}$$

Alternatively, the fitness can be based on minimizing prediction error:

$$\text{Fitness} = 1 / (\text{Prediction Error} + \epsilon)$$

A candidate that correctly predicts more flood events receives a better fitness score and has a higher probability of being selected for the next generation.

4. Gradient-Based Search Using Backpropagation

Backpropagation is a supervised learning algorithm used to train artificial neural networks. It searches for suitable values of network weights by calculating the gradient of the loss function with respect to each weight.

4.1 Neural Network for Flood Prediction

A neural network may receive rainfall, river water level, soil moisture, temperature, humidity, and historical flood information as inputs. These values pass through one or more hidden layers and produce a flood-risk prediction.

Input Features → Hidden Layers → Output Layer → Flood Risk

4.2 Backpropagation Workflow

8. Forward Propagation: Input data is passed through the network to generate a prediction.
9. Loss Calculation: The predicted flood risk is compared with the actual flood condition.
10. Backward Propagation: The error is propagated backward through the network.
11. Gradient Calculation: The contribution of each weight to the error is calculated.
12. Weight Update: Weights are adjusted using gradient descent.
13. Iteration: The process is repeated for many training cycles or epochs.

4.3 Weight Update

The general gradient descent update rule is:

$$\mathbf{w_new} = \mathbf{w_old} - \eta \partial L / \partial \mathbf{w}$$

Here, η represents the learning rate, L represents the loss function, and $\partial L / \partial \mathbf{w}$ represents the gradient of the loss with respect to the weight. The update moves the model toward a region of lower prediction error.

4.4 Flood Prediction Example

If the model predicts 'No Flood' when the actual condition is 'Flood', the loss is high. Backpropagation identifies how the network weights contributed to this error and adjusts them. After repeated training, the network learns patterns associated with high rainfall, rising river levels, and saturated soil.

5. Comparison of the Two Search Approaches

Criterion	Genetic Algorithm	Backpropagation
Search Space	Searches a population of hypotheses	Searches neural-network weight space
Search Mechanism	Selection, crossover, mutation	Gradient calculation and weight update
Gradient Required	No	Yes
Search Style	Population-based and evolutionary	Continuous numerical optimization
Local Optima	Less likely to become trapped	May become trapped in local minima
Computational Cost	Can be high due to many fitness evaluations	Usually efficient for large neural networks
Representation	Chromosomes, rules, parameters, or models	Weights and biases
Flood Application	Feature selection, rule optimization, parameter search	Direct training of a neural network
Exploration	Strong global exploration	Strong local refinement
Convergence	May be slower and stochastic	Usually faster after a suitable gradient path is found

5.1 Key Conceptual Difference

The fundamental difference is that a Genetic Algorithm does not directly follow a mathematical gradient. It explores a population of possible hypotheses and uses their fitness to guide evolution. Backpropagation uses the slope of the loss function to determine the direction in which the weights should move.

Therefore, Genetic Algorithms provide broad exploration of the hypothesis space, while Backpropagation provides efficient fine-tuning of continuous model parameters.

6. Application to Flood Prediction: Case Study Analysis

6.1 Genetic Algorithm Approach

In the flood prediction case study, a Genetic Algorithm can search for the best combination of input features and model parameters. For example, one candidate may give high importance to rainfall and river level, while another may give greater importance to soil moisture and historical flood frequency.

The fitness of each candidate is evaluated using historical flood records. Candidates that produce accurate predictions are selected. Crossover combines useful characteristics from two candidates, and mutation introduces new combinations that may improve the model.

This approach is useful when the search space is complex, discontinuous, or contains multiple possible model structures. It can also be used to select the most useful environmental features.

6.2 Backpropagation Approach

In the Backpropagation approach, the environmental variables are provided as inputs to a neural network. The network produces a flood-risk prediction. The prediction error is calculated, and the weights are updated repeatedly to reduce the loss.

This approach is highly effective when a sufficiently large dataset is available and the relationship between environmental variables and flood risk is complex and nonlinear.

6.3 Example of a Hybrid Approach

The two approaches can also be combined. A Genetic Algorithm may first select important features or optimize neural-network hyperparameters. Backpropagation can then train the neural network weights.

**Genetic Algorithm → Feature/Parameter Optimization → Neural Network → Backpropagation
→ Flood Prediction**

This hybrid approach combines the global search capability of Genetic Algorithms with the efficient local optimization capability of Backpropagation.

6.4 Recommended Approach

For a practical flood prediction system, Backpropagation is generally suitable for training a neural network because it efficiently updates a large number of continuous weights. Genetic Algorithms are especially useful for feature selection, architecture optimization, hyperparameter tuning, or situations where the search space is irregular.

7. SDG 13 – Climate Action

SDG 13 calls for urgent action to combat climate change and its impacts. Floods are among the important climate-related hazards that can affect communities, infrastructure, agriculture, and ecosystems.

A Machine Learning-based flood prediction system can support climate action by analyzing historical and real-time environmental data and identifying patterns associated with increased flood risk.

Contribution to SDG 13

- **Early Warning:** Predicting flood risk can provide additional time for preparedness and response.
- **Climate Resilience:** Data-driven prediction can help communities prepare for extreme weather events.
- **Disaster Risk Reduction:** Authorities can use risk information for planning and emergency management.
- **Resource Optimization:** Emergency resources can be directed toward areas with higher predicted risk.
- **Evidence-Based Decision Making:** Historical environmental data can support better climate adaptation strategies.

8. Advantages and Limitations

Genetic Algorithms

Advantages: They do not require gradient information, can explore complex search spaces, support global exploration, and can optimize feature subsets and model parameters.

Limitations: They may require many fitness evaluations, can be computationally expensive, and their stochastic nature can lead to variable results.

Backpropagation

Advantages: It is efficient for training neural networks, works well with continuous parameters, and can learn complex nonlinear relationships between environmental variables and flood risk.

Limitations: It depends on gradient information, can be affected by local minima or poor initialization, and may require careful selection of learning rate and other training parameters.

9. Conclusion

Hypothesis space search using Genetic Algorithms and gradient-based search using Backpropagation represent two different approaches to Machine Learning optimization. Genetic Algorithms search through a population of candidate hypotheses using selection, crossover, and mutation, providing strong global exploration. Backpropagation searches the neural-network weight space by calculating gradients and updating weights to minimize prediction error.

For flood prediction modeling, Genetic Algorithms are useful for feature selection, parameter optimization, and complex hypothesis search, while Backpropagation is highly effective for training neural networks to learn nonlinear relationships between environmental variables and flood events. A hybrid model can combine both approaches to improve prediction performance.

By supporting early warning, disaster preparedness, climate resilience, and data-driven environmental decision-making, Machine Learning-based flood prediction contributes to SDG 13 – Climate Action.

References

- Mitchell, T. M., Machine Learning, McGraw-Hill.
- Holland, J. H., Adaptation in Natural and Artificial Systems, University of Michigan Press.
- Rumelhart, D. E., Hinton, G. E., and Williams, R. J., Learning Representations by Back-Propagating Errors.
- Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press.
- United Nations, Sustainable Development Goal 13: Climate Action.